

ANSHUL RASTOGI

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CAREER OBJECTIVE

Motivated Computer Science Engineering student with strong foundations in Full-Stack Development, Databases, API design, and AI-based systems. Seeking a software engineering role to build scalable UIs, automate workflows using AI agents, and contribute to production-grade architecture.

EDUCATION

B.Tech - Computer Science & Engineering

VIT Bhopal University | 2023-2027 | CGPA: 9.12

TECHNICAL SKILLS

Languages: JavaScript, Python, Java

Frontend: HTML5, CSS3, React.js, Vite

Backend & APIs: Node.js, REST APIs

Databases: MySQL, Firebase (Cloud Firestore)

Cloud, AI & Tools: Git, Google Gemini AI, Firebase Auth, AWS

Core Subjects: Data Structures & Algorithms, Operating Systems, Computer Networks

PROJECTS

1. Apex Fit – AI-Powered Athlete Performance Tracker

Technologies: JavaScript (ES6+), HTML5/CSS3, Vite, Node.js (Firebase), REST APIs, Git

- Engineered a modern web application using JavaScript and HTML/CSS, leveraging Vite to ensure fast build times and an optimized responsive UI.
- Integrated the Google Gemini AI REST API to process user biometrics and dynamically generate personalized diet, injury prevention, and mental wellness plans.
- Architected the backend utilizing Firebase (Auth & Cloud Firestore), successfully implementing secure Google OAuth and real-time NoSQL data management.
- Built a dynamic Yo-Yo Test Predictor module that evaluates user endurance metrics and fitness levels.

2. Deep Packet Inspection (DPI) Engine – Stateful Network Firewall

Technologies: Java 11, Concurrency (Multi-threading), java.nio (ByteBuffer), TCP/IP & TLS Protocols, Custom Hashing, Git

- Engineered a high-performance multi-threaded firewall in Java, utilizing a Producer-Consumer architecture and thread-safe queues to process raw .pcap network traffic files with zero thread contention.
- Integrated low-level protocol parsers via java.nio.ByteBuffer to decode Ethernet, IPv4, and TCP/UDP headers, successfully extracting TLS Server Name Indication (SNI) data for advanced traffic filtering.
- Architected a scalable load-balancing pipeline leveraging a custom 5-tuple hashing algorithm, ensuring strict packet-to-thread connection affinity and eliminating concurrency race conditions.
- Built a stateful Connection Tracker module that monitors active network flows and executes O(1) time complexity rule lookups to instantly drop malicious or blacklisted domain traffic.

CODING PROFILE

Solved 100+ DSA problems (Arrays, Recursion, Sorting)

CERTIFICATIONS

- Deep Learning - IIT Ropar (NPTEL) | Elite | 76% | Jan-Apr 2025
- ServiceNow - Virtual Internship Program | May-June 2026
- Marketing Management - IIT Kanpur (NPTEL) | Elite (Top 1%) | 100% | Feb-April 2026